

Assignment 00

Your Name Goes Here

August 22, 2026

Assignment 00

Due: See Date in Moodle

Guidelines

- Follow good programming practices by using descriptive variable names, maintaining appropriate spacing for readability, and adding comments to clarify your code.
- Ensure written responses use third-person, correct spelling, complete sentences, and proper grammar.

Name:

Section:

Date:

Quarto

Quarto enables you to weave together content and executable code into a finished document. To learn more about Quarto see <https://quarto.org>.

Running Code

When you click the **Render** button a document will be generated that includes both content and the output of embedded code. You can embed code like this:

```
[1] 2
```

You can add options to executable code like this:

```
{r}
#| echo: false
2*2
```

```
[1] 4
```

The `echo: false` option disables the printing of code (only output is displayed).

You can label code chunks like this:

```
{r}
#| label: load-packages
#| message: false
#| warning: false

library(readxl)
library(dplyr)
library(ggplot2)
```

The `message: false` and `warning: false` options hide messages and warnings from the rendered output.

Loading Data

```
Rows: 26,237
Columns: 33
$ matched_2021      <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ~
$ new_item_2022     <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
$ menu_item_id      <dbl> 10012, 10013, 10018, 10020, 10023, 10025, 10056~
$ food_category     <chr> "Soup", "Soup", "Soup", "Salads", "Salads", "Sa~
$ restaurant        <chr> "Applebee's", "Applebee's", "Applebee's", "Appl~
$ item_name          <chr> "French Onion Soup", "Tomato Basil Soup", "Chic~
$ item_description   <chr> "French Onion Soup, Soups", "Tomato Basil Soup,~
$ serving_size       <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ serving_size_text  <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ serving_size_unit  <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ serving_size_household <chr> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ calories           <dbl> 370, 220, 280, 130, 1560, 230, 1430, 1450, 180,~
$ total_fat          <dbl> 22, 12, 15, 7, 103, 18, 89, 80, 9, 106, 66, 75,~
```

```

$ saturated_fat      <dbl> 12.0, 4.5, 4.0, 2.5, 17.0, 3.5, 14.0, 21.0, 2.0~
$ trans_fat          <dbl> 0.0, 1.5, 0.0, 0.0, 1.0, 0.0, 1.0, 0.5, 0.0, 2.~
$ cholesterol        <dbl> 55, 25, 35, 10, 65, 15, 150, 120, 20, 200, 85, ~
$ sodium             <chr> "1250", "1270", "930", "230", "1610", "410", "2~
$ carbohydrates      <chr> "26", "22", "26", "14", "120", "13", "104", "12~
$ dietary_fiber      <dbl> 2, 2, 2, 2, 12, 2, 11, 9, 1, 8, 7, 9, 1, 2, 7, ~
$ sugar              <dbl> 9, 9, 3, 4, 44, 3, 52, 21, 5, 6, 7, 6, 0, 10, 1~
$ protein            <chr> "16", "5", "11", "6", "40", "5", "56", "56", "7~
$ potassium          <dbl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ notes              <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ calories_text      <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ total_fat_text     <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ saturated_fat_text <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ trans_fat_text     <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ cholesterol_text   <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ sodium_text        <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ carbohydrates_text <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ dietary_fiber_text <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ sugar_text         <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~
$ protein_text       <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, ~

```

[MenuStat](#) is a free nutrition database that aggregates menu and nutrition information from major U.S. chain restaurants, allowing users to compare food items across restaurants and analyze changes in nutritional content over time.

To learn more about the dataset and its variables, review the [Menu Stat Datasheet](#).

```

# A tibble: 6 x 33
  matched_2021 new_item_2022 menu_item_id food_category restaurant item_name
      <dbl>         <dbl>      <dbl> <chr>          <chr>      <chr>
1           1           0      10012 Soup          Applebee's French Onion~
2           1           0      10013 Soup          Applebee's Tomato Basil~
3           1           0      10018 Soup          Applebee's Chicken Tort~
4           1           0      10020 Salads        Applebee's House Salad
5           1           0      10023 Salads        Applebee's Oriental Chi~
6           1           0      10025 Salads        Applebee's Caesar Salad

# i 27 more variables: item_description <chr>, serving_size <chr>,
#   serving_size_text <lgl>, serving_size_unit <chr>,
#   serving_size_household <chr>, calories <dbl>, total_fat <dbl>,
#   saturated_fat <dbl>, trans_fat <dbl>, cholesterol <dbl>, sodium <chr>,
#   carbohydrates <chr>, dietary_fiber <dbl>, sugar <dbl>, protein <chr>,
#   potassium <dbl>, notes <lgl>, calories_text <lgl>, total_fat_text <lgl>,
#   saturated_fat_text <lgl>, trans_fat_text <lgl>, cholesterol_text <lgl>, ...

```

Question 1. Find the 10 restaurants with the fewest menu items in the dataset.

```
# A tibble: 10 x 2
  restaurant      n
  <chr>          <int>
1 Carls Jr       105
2 Sbarro         104
3 The Capital Grille 100
4 Wingstop       96
5 In-N-Out       89
6 Little Caesar's 85
7 Popeyes        79
8 El Pollo Loco   77
9 Five Guys       36
10 Raising Cane's Chicken Fingers 25
```

Question 2. Compute summary statistics for the `calories` variable.

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NAs
0.0	110.0	240.0	387.9	490.0	13734.0	456

Question 3. Create a histogram showing the distribution of calories for all menu items across restaurants. Mark the maximum calorie value on the histogram.

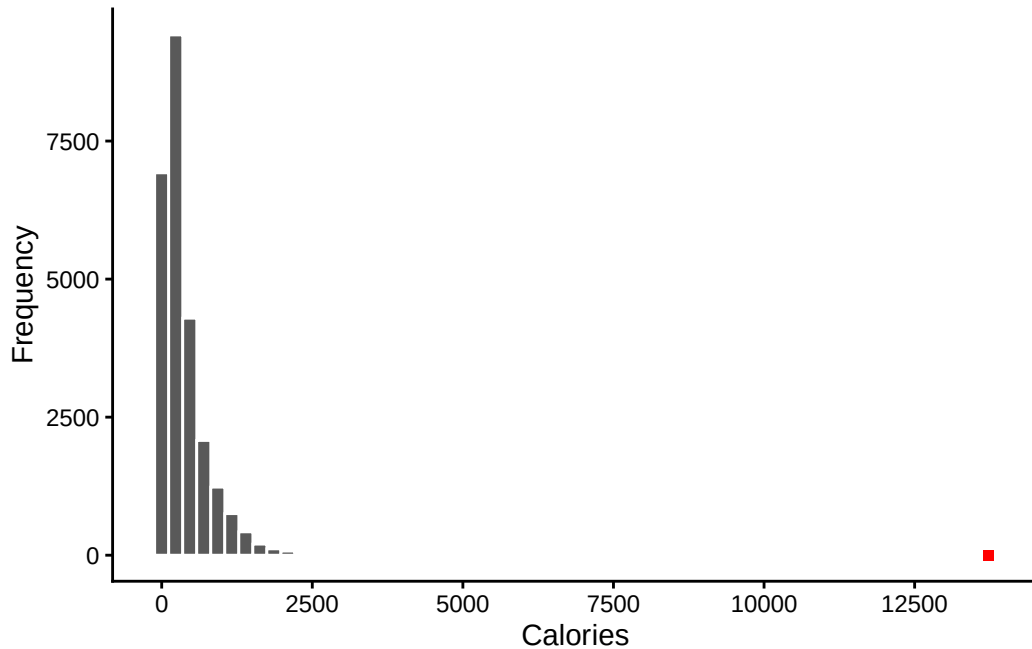


Figure 1: Calorie distribution of all menu items across restaurants.

Question 4. Find the menu item with the highest number of calories.

```
# A tibble: 1 x 4
  restaurant      item_name      item_description      calories
  <chr>          <chr>          <chr>          <dbl>
1 Dickey's Barbeque Party Pack, 12 Servings Party Pack (12 Servings) w~ 13734
```

Question 5. Find the highest-calorie menu item for each restaurant identified in **Question 1**.

```
# A tibble: 10 x 3
# Groups:   restaurant [10]
  restaurant      item_name      calories
  <chr>          <chr>          <dbl>
1 Sbarro        Stuffed Sausage and Pepperoni Pizza 7200
2 Little Caesar's 5 Meat Feast, Detroit-Style Deep Dis~ 3500
3 The Capital Grille Roasted Chicken Breast w/Parmesan Ri~ 2620
4 Raising Cane's Chicken Fingers Caniac Combo 1790
5 Carls Jr      Triple Cheeseburger & Spicy Chicken ~ 1430
6 Five Guys     Fries, Large 1310
7 Popeyes       Chicken Livers 1190
```

8 Wingstop	Louisiana Voodoo Fries Large	1180
9 El Pollo Loco	Chips & Queso (regular)	1080
10 In-N-Out	Double-double w/ Onion	670

Question 6. Create a bar chart showing the highest-calorie menu item for each restaurant identified in **Question 1**.

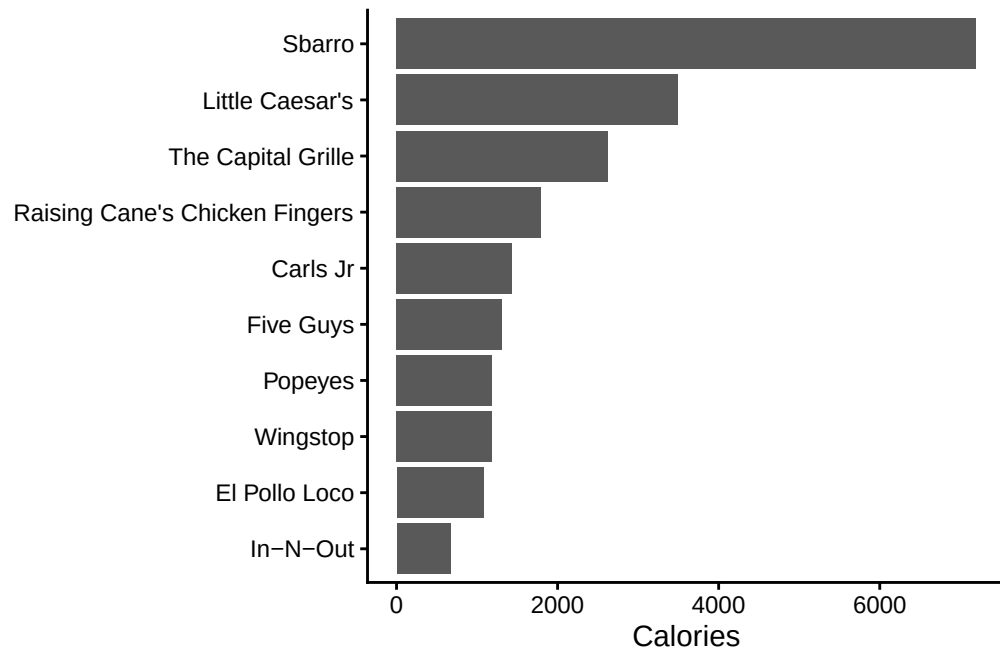


Figure 2: Highest-calorie menu item for selected restaurants.

Question 7. Create a boxplot showing the calorie distribution for menu items from the top five restaurants identified in **Question 1**.

Question 8. Identify and inspect invalid or non-numeric values in the `sodium` variable.

```
[1] "160+"      "0+"      NA        "25-50"    "3410m"
[6] "5650m"    "-"      "0-35"    "1760\r\n\r\n" "2, 280"
[11] "1, 240"    "1, 230"  "2, 290"  "1, 290"    "3, 200"
[16] "4, 280"    "90-140"
```

```

-      0-35      0+      1, 230      1, 240      1, 290
4      1      2      1      1      1
160+ 1760\r\n\r\n  2, 280  2, 290  25-50  3, 200
1      1      1      1      1      1
```

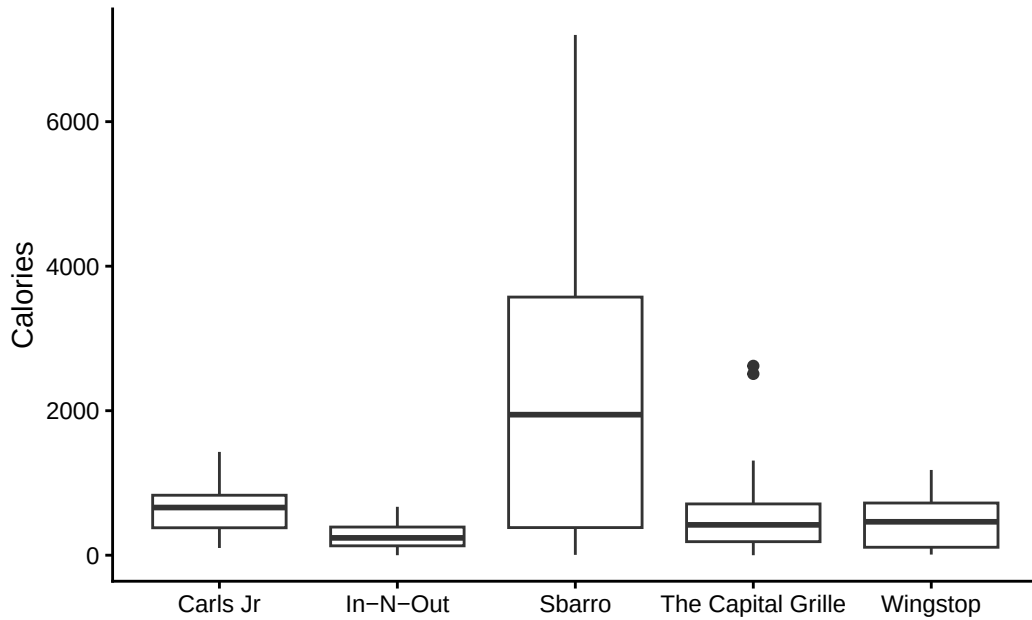


Figure 3: Calorie distribution of menu items for selected restaurants.

3410m	4, 280	5650m	90-140	<NA>
1	1	1	1	965

Question 9. Clean and summarize the `sodium` variable by keeping only numeric values.

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0	50.0	260.0	748.1	930.0	49051.0

Question 10. Clean and summarize the `sodium` variable by keeping only numeric values. Then create a boxplot showing the calorie distribution for menu items from the top five restaurants identified in **Question 1**.

Submission

Make sure that all cells in your assignment have been executed to display all output, images, and graphs in the final document.

Note: Save the assignment before proceeding to render the file into a `.pdf` document.

Once the rendering process is complete, locate the resulting `.pdf` document and upload **only** this file to Moodle. The assignment will be automatically submitted to Gradescope for grading.

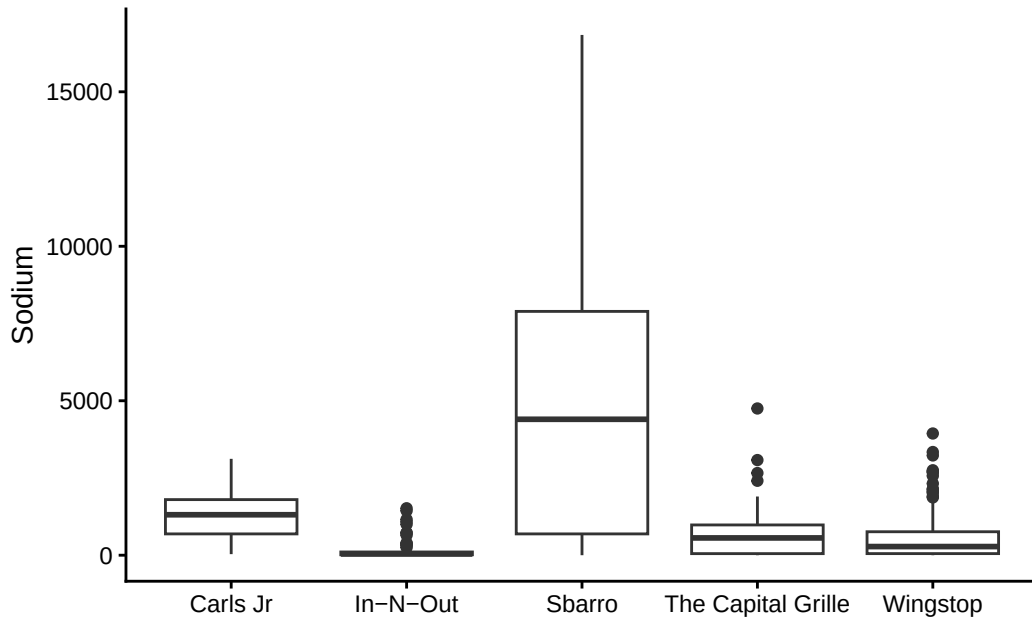


Figure 4: Sodium distribution of menu items for selected restaurants.

If you have any questions, please post them on the Moodle discussion board dedicated to this assignment.

Grading Rubric

This programming assignment is worth **15 points**: 10 points for the questions, 4 points for submission compliance, and 1 point for timeliness. See the [Grading Rubric Guideline](#) for details on marking tags and point values.